

GetRightHome Home Services

Proposed Booking / Slot / Auto-Assignment Flow vs Current Codebase: Gap Analysis

Source: backend/ (hsBookingController, waveScheduler, locationService, workerBookingController, adminWorkerController, Worker & HomeServiceBooking models, Settings) and frontend/src/homster. Generated 15 Sep 2026 from repo state at commit 7e725ab.

MAJOR GAP = redesign / naya module

PARTIAL = concept hai, spec se alag

MATCH = already spec jaisa

#	Area	Spec (naya flow)	Current code	Gap
1	Core model	Customer chooses WHEN, worker declares availability, system decides WHO (auto-assign)	Broadcast/wave model: booking bante hi 3 nearest online workers ko request jaati hai, worker ko accept karna padta hai (waveScheduler.js, locationService.js)	MAJOR GAP Fundamental difference. Auto-assign nahi hai, "first to accept" hai
2	Worker online/offline	Online status future booking eligibility decide na kare	Matching query me isOnline: true hard filter hai. Offline worker kabhi consider nahi hota	MAJOR GAP Spec Section 6 (offline worker ko assign) bilkul support nahi
3	Worker availability	Worker + service + date + time slot + area ke against availability store ho	Worker model me sirf isOnline, status (online/offline/busy), serviceCategories, geoLocation. Koi availability calendar/slot schema nahi	MAJOR GAP Poora naya model chahiye (WorkerAvailability) + worker app me "My Availability" screen
4	Matching priority rules	1 skill, 2 exact slot, 3 distance, 4 workload/overlap, 5 rating, 6 reliability, 7 admin rules	Sirf: zone polygon + \$geoNear distance, approved+active+online, category ka loose substring match (match na mile to sab nearby workers ko bhej deta hai). Rating, workload, slot overlap, reliability, admin rules nahi	MAJOR GAP 7 me se sirf ~1.5 rules implemented (skill loosely, distance)
5	Primary + backup worker	System internally primary aur backup pool rakhe	potentialWorkers[] distance-sorted list hai, lekin wo wave batches ke liye hai, backup concept nahi	MAJOR GAP Backup pool nahi hai
6	Booking order (pay -> confirm -> assign)	Payment complete hone ke baad hi booking confirm aur matching start ho	createBooking booking ko paymentStatus: pending ke saath save karta hai aur setImmediate me turant wave 1 dispatch kar deta hai. Razorpay order booking create hone ke baad banta hai; actual settle aksar work_done -> awaiting_payment par hota hai	MAJOR GAP Dispatch payment se pehle ho jaata hai. Unpaid booking bhi workers ko chali jaati hai
7	Payment rules	<= ₹2000 full payment, > ₹2000 par 30% advance, dono admin-configurable	Aisa koi rule nahi. Sirf estimate-based jobs me 30% token hardcoded hai (workerBookingController.js:1471). Settings/PlatformSettings me threshold ya advance % field nahi	MAJOR GAP Poora rule + settings fields missing
8	Slot reservation during payment	Slot Temporarily Reserved -> Booked / wapas Available	Slot ka concept hi nahi. timeSlot {start,end} sirf booking par store hota hai, kahin lock/reserve nahi hota	MAJOR GAP Reservation + expiry mechanism nahi
9	Double booking prevention	Confirmed booking worker ka slot lock kare	Worker accept karte waqt sirf yeh check hai ki "yeh booking kisi aur ne le li?" Us worker ki overlapping bookings check nahi hoti	MAJOR GAP Ek worker same slot me multiple jobs accept kar sakta hai
10	Worker "Can't Attend" / reassignment	Reason -> WORKER_REASSIGNMENT -> backup pool se next worker auto-assign -> customer update -> koi na mile to admin	releaseJob hai: reason assignmentAttempts me save, assignmentStatus: 'reassigning', status wapas searching, wave restart, dropped worker exclude. work_done ke baad block	PARTIAL Sabse close match. Fark: backup se auto-assign nahi, dbara broadcast hota hai
11	No worker available	Fail mat karo; "Finding a Professional / Admin Action Required". Admin manually assign, slot/date change, ya contact kare	manual_assignment_required status, cancel nahi hota, POST /admin/workers/jobs/:id/assign se manual assign, stale escalation cron	MATCH Match. Sirf "admin offers another slot/date" nahi hai (reschedule sirf user kar sakta hai)

#	Area	Spec (naya flow)	Current code	Gap
12	Status lifecycle	DRAFT -> PAYMENT_PENDING -> PAYMENT_COMPLETED -> CONFIRMED -> WORKER_ASSIGNMENT_PENDING -> WORKER_ASSIGNED -> ACCEPTED -> ON_THE_WAY -> ARRIVED -> SERVICE_STARTED -> SERVICE_COMPLETED -> CUSTOMER_CONFIRMED -> COMPLETED	searching -> assigned -> journey_started -> visited -> in_progress -> work_done -> awaiting_payment -> completed, plus estimate_provided/accepted, manual_assignment_required, legacy no_workers/no_vendors/confirmed/accepted. Payment alag paymentStatus field me	PARTIAL Naming/order alag, mapping possible (ON_THE_WAY = journey_started, ARRIVED = visited, etc.). Missing: PAYMENT_PENDING/COMPLETED as booking states, REFUND_PENDING. Extra: estimate states. updateJobStatus loose table assigned -> completed direct allow karti hai
13	Exception statuses	WORKER_REASSIGNMENT, CANCELLED, REFUND_PENDING, REFUNDED	assignmentStatus: 'reassigning', cancelled, paymentStatus: 'refunded'. Refund user ke in-app wallet me jaata hai, Razorpay refund nahi	PARTIAL REFUND_PENDING nahi; refund gateway tak nahi jaata
14	Notifications matrix	Customer / worker / admin ko har status event par	Customer aur worker ke liye zyadatar events hain (Booking Received, Assigning, Worker Assigned, Started Journey, Reached, Work Done, Cancelled, Declined). Admin ko sirf admin_room socket kuch events par	PARTIAL Mostly covered. Admin-side "no worker available" aur "worker cancellation" alerts confirm karne honge
15	Worker panel	My Schedule, My Availability, Notifications, Job Details, Profile	AssignedJobs, JobDetails, JobMap, JobTimeline, Notifications, Profile, Documents, Wallet, Subscription, Training, Referrals	PARTIAL "My Availability" nahi hai. "My Schedule" ~ AssignedJobs (calendar view nahi)
16	Worker ko financials na dikhe (API level)	Customer total, commission, margin, payout calc, advance: backend par restrict	UI level par hide kiya gaya (recent commits). Backend getJobById me finalAmount saath me workerAmount bhejta hai, VendorBill me adminCommission return hota hai	MAJOR GAP Sirf UI hiding hai, API restriction nahi
17	Admin filters	date, category, area, worker, status, payment status, unassigned, pending acceptance, cancelled, completed	getAllJobs: sirf status, search, startDate, endDate	PARTIAL category, area/zone, worker, paymentStatus, unassigned, pending-acceptance filters missing
18	Audit trail	Assignment, reassignment, cancellation ka trail	assignmentAttempts[] (workerId, outcome, reason, timestamps) + cancellationReason. General statusHistory nahi	PARTIAL Partial. Status change history nahi
19	Config in Admin Settings	Payment threshold, advance % configurable	Settings me waveDuration, searchRadius, cancellationPenalty hain; payment threshold/advance nahi	PARTIAL Fields add karne honge
20	Service area	Customer address service area me ho	Zone polygon check booking se pehle, out-of-zone par OUT_OF_SERVICE_AREA	MATCH Match
21	Server-side pricing	System price calculate kare	Server trusted Service records se re-price karta hai	MATCH Match
22	Customer confirmation on completion	CUSTOMER_CONFIRMED step	Customer OTP (customerConfirmationOTP) close-out par	MATCH Match (naam alag)

Summary

Jo already spec se match karta hai: zone-based service area, server-side pricing, worker can't-attend with reason, no-worker case admin queue me jaana (cancel nahi), customer OTP confirmation, aur customer/worker notifications ka bada hissa. Status lifecycle bhi rename/remap se kaam chal sakta hai.

Sabse bade gaps, priority order me:

1. **Matching model hi alag hai.** Current system "online workers ko broadcast, jo pehle accept kare" hai. Spec "availability calendar se auto-assign" maangta hai. Yeh locationService, waveScheduler aur worker accept flow ka redesign hai, patch nahi.
2. **Worker availability data model exist nahi karta.** Bina iske slot matching, overlap check, offline assignment, slot lock kuch bhi possible nahi.
3. **Payment-before-dispatch nahi hai.** Booking create hote hi dispatch shuru ho jaata hai, payment baad me. Spec me payment gate pehle hai, plus 30% advance rule aur slot reservation.
4. **Financial restriction API level par nahi.** Abhi finalAmount aur adminCommission worker endpoints se nikal rahe hain.
5. **Admin filters aur audit trail** chhote gaps hain, additive kaam hai.

Note: current code me pay_at_home fallback abhi bhi checkout me default hai jabki backend cash hata chuka hai. Naye flow me yeh conflict pehle resolve karna padega kyunki "payment first" isi par depend karta hai.

Status mapping (spec -> current)

Spec status	Current equivalent	Note
DRAFT	(none)	Booking create se pehle koi persisted draft nahi
PAYMENT_PENDING / PAYMENT_COMPLETED	paymentStatus: pending / paid	Alag field, booking status nahi
CONFIRMED	confirmed (legacy) / searching	Payment ke baad confirm nahi hota; create hote hi searching
WORKER_ASSIGNMENT_PENDING	searching (+ assignmentStatus: searching)	Wave broadcast chal raha hota hai
WORKER_ASSIGNED	assigned	Worker accept karne par
WORKER_ACCEPTED / CONFIRMED	assigned (workerResponse: ACCEPTED)	Assign aur accept ek hi step hai
ON_THE_WAY	journey_started	visitOtp issue hota hai
ARRIVED	visited	visitOtp verify
SERVICE_STARTED	in_progress	Estimate jobs me token pay ke baad
SERVICE_COMPLETED	work_done -> awaiting_payment	Bill yahan banta hai
CUSTOMER_CONFIRMED	customerConfirmationOTP verify	Status nahi, OTP step
COMPLETED	completed	Match
WORKER_REASSIGNMENT	assignmentStatus: reassigning + status: searching	releaseJob
CANCELLED	cancelled	Match; refund wallet me
REFUND_PENDING / REFUNDED	paymentStatus: refunded	REFUND_PENDING nahi
(no equivalent)	manual_assignment_required, estimate_provided, estimate_accepted, no_workers, no_vendors, rejected	Extra current states